AI CAREER FOR WOMEN

Engineering Hub



AI Based Scheme Recommendation and Eligibility System

A Project Report

submitted in partial fulfillment of the requirements

of

AICW project.

by

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ABSTRACT of the Project

The **AI-Based Government Scheme Recommendation and Eligibility Assistance System** is an intelligent system designed to help individuals identify government welfare schemes that may be relevant to their socioeconomic conditions. Many people are unaware of available government schemes, their eligibility requirements, and the documents or procedures required to access benefits. This creates difficulties in discovering suitable schemes and may lead to underutilization of available welfare programmes.

The main objective of this project is to develop a personalized system that analyzes user socioeconomic information and recommends potentially relevant government schemes. The system also aims to provide understandable explanations for recommendations, identify possible benefit-access gaps, and guide users toward the required documents and next steps.

The proposed methodology involves **data preprocessing, exploratory data analysis (EDA), feature engineering, machine learning-based pattern analysis, and a government-scheme knowledge base**. A hybrid matching approach combines user profile information with scheme eligibility criteria to generate personalized recommendations. The system further provides a **"Why Suggested?"** explanation for each recommendation and presents eligibility assistance rather than making a final eligibility decision.

The key result of the proposed system is a structured and personalized scheme-discovery process that can reduce the difficulty of finding relevant welfare programmes and improve users' understanding of available benefits. The system is designed to present recommendations clearly while directing users to official sources for final eligibility verification.

In conclusion, the proposed system demonstrates how **AI, machine learning, and data analytics** can be applied to improve government scheme discovery and provide accessible, explainable, and personalized eligibility assistance.

TABLE OF CONTENTS

Abstract ----------

List of Figures ----------

List of Tables ----------

Chapter 1. Introduction

1.1 Problem Statement

1.2 Motivation

1.3 Objectives

1.4. Scope of the Project

Chapter 2. Literature Survey

Chapter 3. Proposed Methodology

Chapter 4. Implementation and Results

Chapter 5. Discussion and Conclusion

References ----------

LIST OF FIGURES

S.NO	Figure Name	Page No
1.	Problem Statement	1
2.	Introduction to the Proposed Government Scheme Recommendation System	3
3.	System Design	9
4.	Personalized Government Scheme Discovery and Eligibility Assistance	12

LIST OF TABLES

S.NO	Table Name	Page No
1.	Dataset Attributes and Description	17
2.	Derived Features	18
3.	Government Scheme Knowledge Base	19
4.	Benefit-Gap Analysis	20

CHAPTER 1

INTRODUCTION

1.1 Problem Statement

Government welfare schemes are designed to support people based on different socioeconomic conditions such as income, age, occupation, education, location, family characteristics, and other eligibility factors. However, many individuals are not aware of the schemes available to them or do not know which schemes may be relevant to their circumstances. The large number of schemes and different eligibility requirements make it difficult for users to identify suitable programmes.

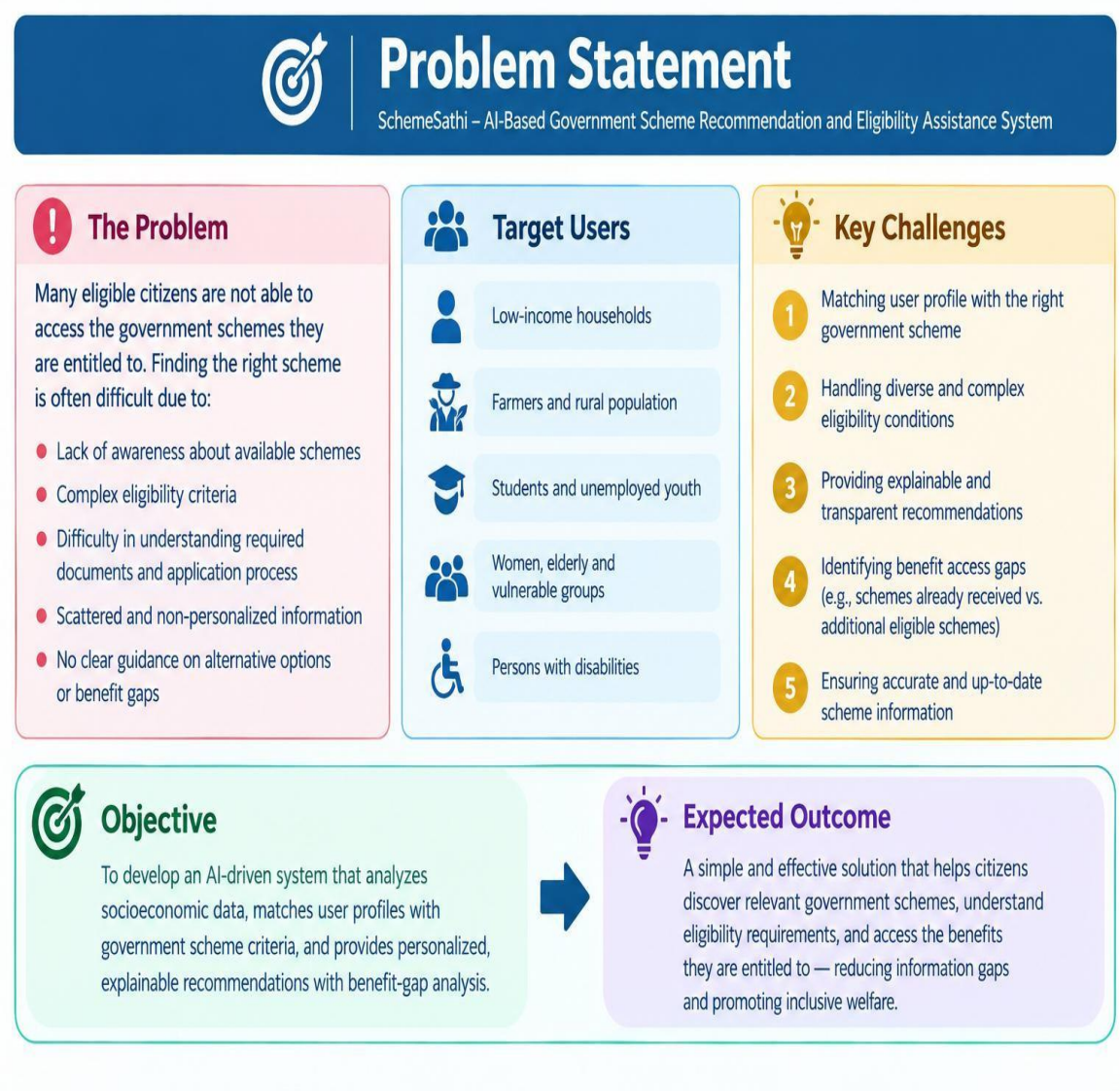


Fig 1:Problem Statement

The existing process often requires people to search through multiple government websites and understand complex eligibility conditions manually. This may result in difficulty in discovering relevant benefits and possible underutilization of welfare programmes.

The proposed **AI-Based Government Scheme Recommendation and Eligibility Assistance System (SchemeSathi)** addresses this problem by analyzing socioeconomic information and identifying government schemes that may be relevant to a user's profile. The system also explains why a scheme was recommended and provides guidance about information or documents that should be verified.

The system is intended as an **eligibility-assistance and discovery tool**, not as an authority for determining final legal eligibility. Actual eligibility must be verified using the latest official government rules and requirements.

1.2 Motivation

The main motivation for developing SchemeSathi is to make government welfare-scheme discovery easier, more personalized, and understandable. Instead of requiring users to manually search through a large number of schemes, the system uses their socioeconomic profile to identify potentially relevant programmes.

The project was chosen because it combines **Artificial Intelligence, Machine Learning, Data Analytics, and Explainable AI** to address a practical social problem. The system can analyze factors such as age, income, occupation, education, family size, employment status, location, and other available socioeconomic characteristics.

Potential applications include:

- Personalized government scheme discovery.
- Eligibility assistance for welfare programmes.
- Identification of possible benefit-access gaps.
- Explanation of why a particular scheme was suggested.
- Guidance about documents and verification steps.
- Simple presentation of complex scheme information.
- Future multilingual support for improved accessibility.

The expected impact is to reduce the difficulty of finding relevant welfare programmes and help users understand which schemes they should investigate further.

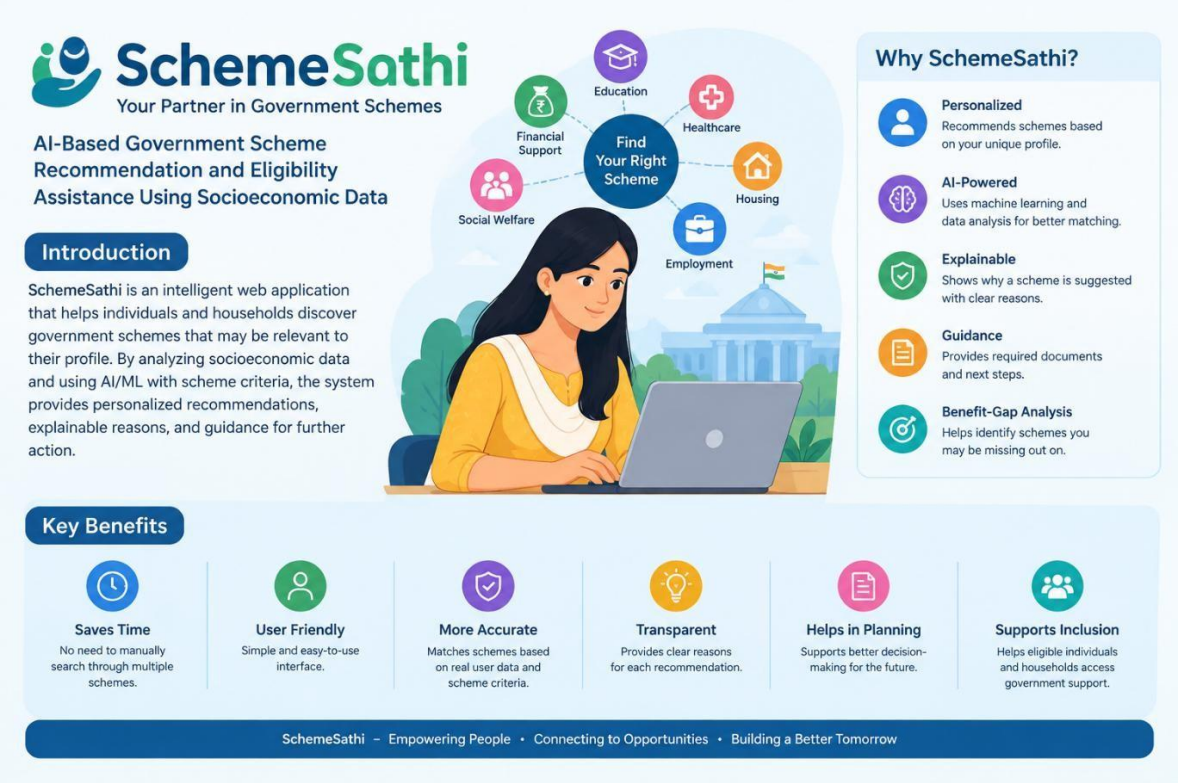


Fig 2: Introduction to the Proposed Government Scheme Recommendation System

1.3 Objectives

The main objectives of the project are:

1. **To develop an AI-based system** for personalized government scheme recommendation.
2. **To analyze socioeconomic information** such as age, income, occupation, education, family characteristics, and location.
3. **To identify potentially relevant government schemes** by comparing user profiles with scheme criteria.
4. **To apply machine learning and data analytics techniques** for profile analysis and recommendation support.
5. **To provide explainable recommendations** by showing why a particular scheme was suggested.
6. **To identify possible benefit-access gaps** where the available data supports such analysis.
7. **To provide document and next-step guidance** for further verification.

8. **To present the results through a user-friendly interface** that makes scheme information easier to understand.
9. **To ensure that recommendations are treated as assistance**, while final eligibility is verified through official government sources.

1.4 Scope of the Project

The scope of SchemeSathi covers the development of a data-driven system that analyzes socioeconomic profiles and provides personalized government-scheme discovery and eligibility assistance.

The project includes:

- Collection and preparation of suitable socioeconomic data.
- Data cleaning and preprocessing.
- Exploratory Data Analysis (EDA).
- Feature engineering and household/profile analysis.
- Machine learning-based pattern analysis.
- A government-scheme criteria knowledge base.
- Rule-based and ML-assisted scheme matching.
- Relevance ranking of potentially suitable schemes.
- Explanation of recommendation factors.
- Benefit-gap analysis where appropriate data is available.
- Document and next-step guidance.
- Development of a simple web-based interface.

The project is limited by the quality, availability, and completeness of the dataset and scheme information. Government scheme rules and eligibility conditions can change over time; therefore, the system cannot guarantee final eligibility. Recommendations should be considered **potentially relevant suggestions**, and users must verify the latest eligibility requirements, documents, and application procedures through official sources.

The project also does not replace government verification or application authorities. Its primary purpose is to support **scheme discovery, profile-based recommendation, explanation, and eligibility assistance**.

CHAPTER 2

LITERATURE SURVEY

2.1 Review of Relevant Literature and Previous Work

Government welfare programmes are designed to provide financial, social, educational, employment, healthcare, housing, and other forms of support to eligible individuals and households. However, identifying suitable welfare schemes can be difficult because different programmes have different eligibility conditions, documentation requirements, and target groups. Traditional approaches to welfare-scheme discovery mainly focus on providing scheme information through databases and eligibility-rule systems, where users are required to manually search for suitable programmes and compare their eligibility conditions. Such approaches may require considerable effort, particularly when users are unfamiliar with government terminology or when a large number of schemes are available.

Recent developments in Artificial Intelligence (AI), Machine Learning (ML), and data analytics provide opportunities to improve welfare-scheme discovery through personalized analysis. Socioeconomic characteristics such as age, income, occupation, education, family size, employment status, social category, disability, and location can be analyzed to identify patterns within user profiles. Such analysis can support the identification of government programmes that may be relevant to a particular individual or household. The reviewed project literature also emphasizes the importance of maintaining a separate government-scheme criteria knowledge base containing information about scheme conditions and target groups. This is useful when the available socioeconomic dataset does not directly contain scheme-eligibility labels.

Another important area in intelligent recommendation systems is explainability. A system that only displays the name of a recommended scheme may not provide sufficient information to the user about why that scheme was suggested. Explainable recommendation can provide understandable reasons based on profile characteristics, such as income, occupation, age, education, or location. This improves transparency and helps users understand the factors associated with a recommendation. Therefore, the reviewed literature and project analysis support the development of a hybrid approach that combines socioeconomic data analysis, machine learning or pattern analysis, rule-based scheme criteria, recommendation, and explanation.

2.2 Existing Models, Techniques and Methodologies

Several models, techniques, and methodologies are relevant to the development of the proposed SchemeSathi system. One important technique is rule-based eligibility matching, in which the user's profile is compared with predefined criteria associated with government schemes. Information such as income, age, occupation, location, education, and other socioeconomic characteristics can be matched against the corresponding scheme conditions. This approach is straightforward and useful when scheme conditions are clearly defined. However, its

effectiveness depends on the accuracy, completeness, and regular maintenance of the predefined rules and scheme information.

Machine Learning-based analysis can also be used to identify patterns within socioeconomic data and support personalized scheme recommendation. Depending on the final dataset and the availability of suitable target information, classification or recommendation techniques can be considered. The performance of an ML model can be evaluated using measures such as Accuracy, Precision, Recall, F1-score, and Confusion Matrix. For recommendation or ranking-based systems, suitable recommendation and ranking metrics can also be considered. The exact machine learning algorithm should be selected after analyzing the characteristics of the final dataset rather than assuming a particular model in advance.

Exploratory Data Analysis (EDA) is another important methodology used to understand the characteristics and relationships present in socioeconomic data. Variables such as income, age, occupation, education, employment status, and family size can be analyzed using statistical summaries and visualizations. EDA helps identify data distributions, missing values, unusual observations, and relationships between variables before applying machine learning or recommendation techniques. Tools such as Pandas, NumPy, Matplotlib, and Plotly can be used for data preprocessing, statistical analysis, and visualization.

A hybrid recommendation approach combines data-driven analysis with predefined government-scheme criteria. In this methodology, socioeconomic data is first collected and preprocessed, followed by EDA and ML or pattern analysis. The resulting user profile is then compared with the government-scheme knowledge base through a matching engine. The matched schemes can subsequently be ranked according to their relevance, and explanations can be generated for the recommendations. The general methodology can therefore be represented as socioeconomic dataset, data preprocessing, ML or pattern analysis, scheme rules, hybrid matching, relevance ranking, explanation, and recommendation.

Explainable recommendation is an additional component of the proposed methodology. Instead of simply displaying a scheme name, the system can provide understandable information about the factors associated with the recommendation. For example, the system may indicate that the user's income condition appears compatible with a particular scheme, the occupation matches the intended target group, an age-related condition appears relevant, or the location condition appears compatible. Such explanations can make the recommendation process more transparent and easier for users to understand.

2.3 Research Gaps and Limitations of Existing Solutions

The review of existing approaches identifies several research gaps and limitations. One major limitation is the requirement for manual scheme search. Users may need to search through multiple government programmes and understand different eligibility requirements individually. This process can be time-consuming and difficult, especially for users who are unfamiliar with government terminology. SchemeSathi addresses this limitation by analyzing the user's socioeconomic profile and presenting schemes that are potentially relevant to the provided information.

Another limitation is the lack of personalization in general-purpose government scheme information systems. A common list of welfare programmes does not necessarily indicate which schemes may be relevant to a particular individual or household. The proposed system addresses this gap by using socioeconomic characteristics to perform personalized scheme matching and recommendation.

Limited explanation is another important gap. A recommendation without a clear explanation may not help users understand why a particular programme was suggested. SchemeSathi therefore incorporates a "Why Suggested?" explanation layer that presents the profile factors associated with the recommendation. This can include factors such as income, occupation, age, education, family characteristics, or location, depending on the available information and scheme criteria.

Existing solutions may also depend primarily on a single methodology. A purely rule-based system can directly represent predefined scheme conditions but may not utilize patterns present in socioeconomic data. On the other hand, a purely machine-learning-based system may not directly represent current government eligibility requirements unless suitable labelled data is available. The proposed system therefore follows a hybrid approach that combines machine learning or socioeconomic pattern analysis with a government-scheme criteria knowledge base.

Another research gap is the limited consideration of possible benefit-access gaps. Basic scheme-discovery systems may only display available programmes without considering whether potentially relevant benefits are already recorded as being received. Where suitable information is available, SchemeSathi can analyze the profile and existing benefit information to identify possible benefit-access gaps and indicate programmes that may be worth checking.

Eligibility interpretation is also an important limitation that must be considered in an automated welfare recommendation system. Government rules, documentation requirements, and verification procedures may change over time. Therefore, an automated system should not claim that a user is legally or finally eligible for a particular scheme. SchemeSathi uses terms such as "potentially relevant" and "may be eligible" and guides users to verify final eligibility using the current official requirements and application procedures.

Overall, the identified literature gaps support the development of an integrated system that combines socioeconomic data, data preprocessing, exploratory data analysis, machine learning or pattern analysis, a government-scheme knowledge base, hybrid scheme matching, relevance ranking, explainable recommendation, benefit-gap analysis, and document or next-step guidance. Thus, SchemeSathi is designed as a personalized, explainable, and data-driven government scheme discovery and eligibility-assistance system rather than only a conventional scheme information portal.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 System Design

The proposed **SchemeSathi – AI-Based Government Scheme Recommendation and Eligibility Assistance System** is designed to analyze the socioeconomic information of individuals or households and identify government welfare schemes that may be relevant to their profile. The system collects user information such as age, income, occupation, education, employment status, family size, location, and other relevant socioeconomic attributes. The collected information is preprocessed and used to create a socioeconomic profile. This profile is analyzed using exploratory data analysis, feature engineering, machine learning or pattern-analysis techniques, and a rule-based government scheme knowledge base. The system then performs hybrid scheme matching, ranks potentially relevant schemes, explains the reasons for recommendations, identifies possible benefit-access gaps, and provides document and next-step guidance. The system provides potentially relevant recommendations and does not make a final legal eligibility decision. Actual eligibility must be verified using the latest official government rules and requirements.

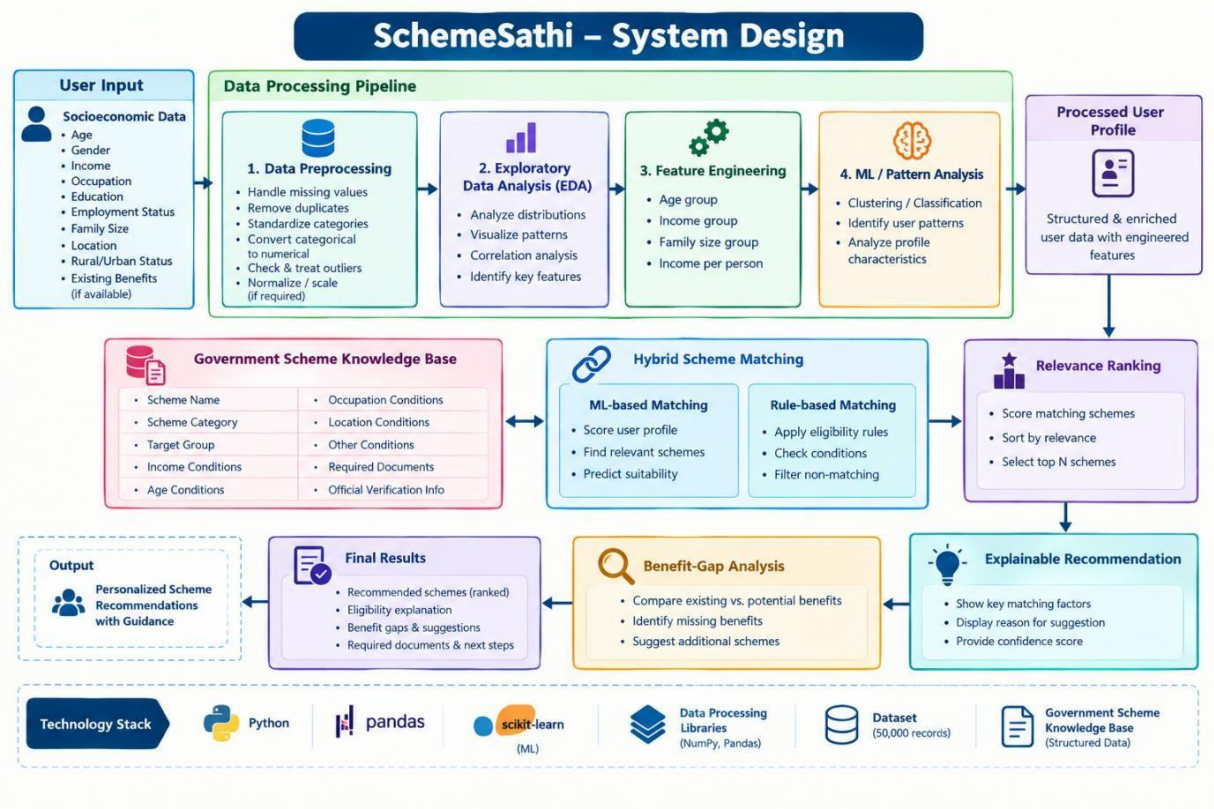


Fig 3:System Design

3.1.1 Registration

The registration module allows users to create an account and provide the basic information required to access the SchemeSathi system. The user may provide details such as name, contact information, and other required profile information. After registration, the user can enter or update socioeconomic details that are required for scheme recommendation. The registration process helps maintain the user's profile and allows the system to use the provided information during subsequent scheme analysis.

3.1.2 Socioeconomic Profile Creation

The socioeconomic profile module collects information such as age, gender, income, occupation, education, employment status, family size, location, rural or urban status, and existing benefits where available. This information is used to create a structured profile representing the user's socioeconomic characteristics. The profile acts as the input for the preprocessing, machine learning, rule-based matching, and recommendation modules.

3.2 Modules Used

The proposed system consists of several modules that work together to provide personalized government scheme recommendations. The major modules include data preprocessing, exploratory data analysis, feature engineering, machine learning or pattern analysis, government scheme knowledge base, hybrid scheme matching, relevance ranking, explainable recommendation, benefit-gap analysis, document and next-step guidance, and web application modules.

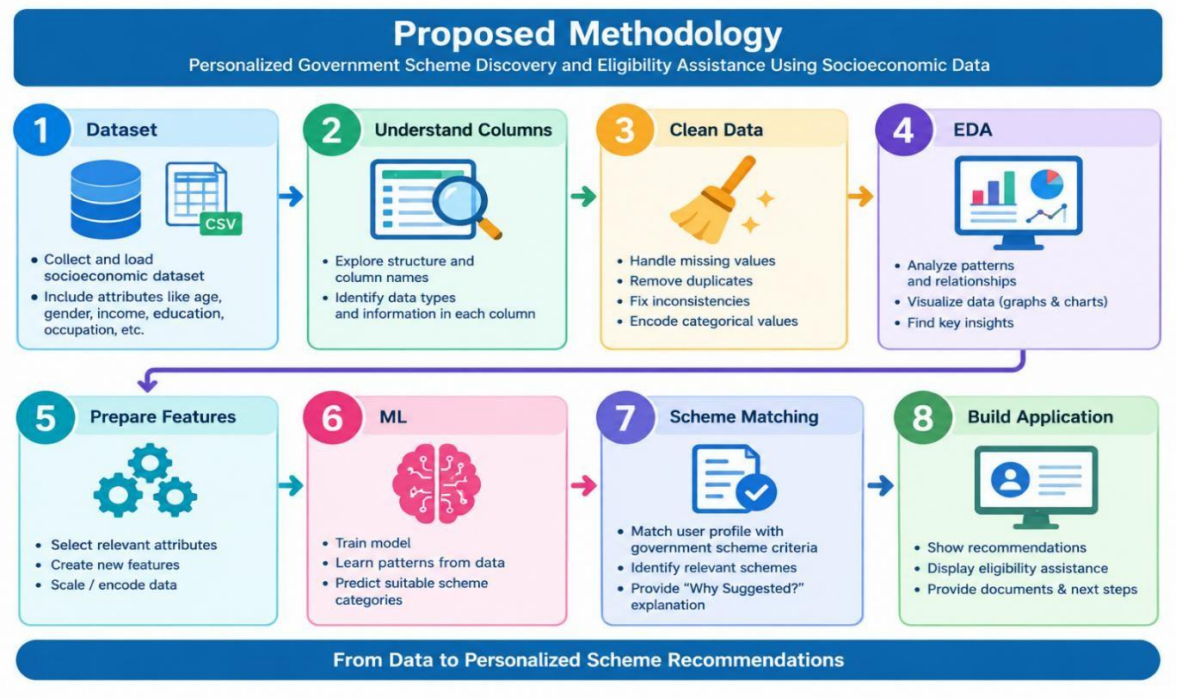


Fig 3: Personalized Government Scheme Discovery and Eligibility Assistance

3.2.1 Data Preprocessing

The data preprocessing module prepares the collected socioeconomic data for further analysis. Missing values are handled, duplicate records are removed, inconsistent values are corrected, and categorical variables are encoded into suitable formats. Numerical attributes are also processed, and scaling or normalization can be applied where required. The resulting cleaned dataset is used for exploratory data analysis, feature engineering, and machine learning.

3.2.2 Exploratory Data Analysis

The EDA module analyzes the socioeconomic dataset to understand important patterns and relationships. Variables such as income, age, occupation, education, employment status, family size, location, and existing benefits can be analyzed using statistical summaries and visualizations. This analysis helps identify important characteristics and relationships that can support the recommendation process.

3.2.3 Feature Engineering

The feature engineering module identifies and prepares relevant attributes for machine learning and scheme matching. Existing attributes may be transformed into useful categories or derived features. For example, income may be grouped into income categories, and household-level information can be transformed into suitable features for analysis. The resulting features are used by the machine learning and recommendation components.

3.2.4 Machine Learning / Pattern Analysis

The machine learning or pattern-analysis module analyzes relationships within the socioeconomic data. Depending on the final dataset and available target variables, classification, pattern identification, scheme-category prediction, or recommendation-support approaches may be used. Possible algorithms include Logistic Regression, Decision Tree, Random Forest, KNN, and Gradient Boosting. The final algorithm will be selected based on the characteristics of the dataset and the target variable. The ML component supports recommendation but does not independently determine legal eligibility.

3.2.5 Government Scheme Knowledge Base

The government scheme knowledge base stores structured information about government welfare schemes and their documented conditions. It may contain scheme name, category, target group, income conditions, age conditions, occupation conditions, location conditions, required documents, and official verification sources. The knowledge base provides the criteria required by the rule engine for scheme matching.

3.2.6 Hybrid Scheme Matching

The hybrid scheme matching module combines machine learning or pattern analysis with rule-based scheme criteria. The user's socioeconomic profile is compared with the conditions stored in the government scheme knowledge base. The rule engine checks configured scheme conditions, while the ML component supports pattern-based analysis. The combined result is used to identify schemes that are potentially relevant to the user.

3.2.7 Relevance Ranking

The relevance ranking module ranks the potentially relevant schemes according to the degree of matching between the user's profile and the scheme criteria. Factors such as the number of matching conditions, importance of individual attributes, profile similarity, and scheme-category relevance can be considered. The ranked results help users understand which schemes may be more relevant to their provided information.

3.2.8 Explainable Recommendation

The explainable recommendation module provides reasons for suggesting a particular scheme. Instead of displaying only the scheme name, the system can explain that the recommendation is based on factors such as income, age, occupation, education, or location. Feature importance or SHAP-based explanations may also be considered for the final ML model where appropriate. This makes the recommendation process more transparent and understandable.

3.2.9 Benefit-Gap Analysis

The benefit-gap analysis module compares potentially relevant schemes with the benefits already recorded for the user or household. If a potentially relevant scheme is not present among the recorded benefits, the system identifies it as a possible benefit-access gap. This indicates that the user may need to check the scheme rather than confirming that the user is definitely eligible.

3.2.10 Document and Next-Step Guidance

This module provides information about the conditions and documents that should be verified before applying for a scheme. The system may display required documents, eligibility conditions to verify, official sources, application information, and suggested next steps. This helps users understand what information is required for further verification.

3.3 Data Flow Diagram

A **Data Flow Diagram (DFD)** is a graphical representation of the flow of data through an information system. It represents how data enters the system, how it is processed, where it is stored, and how the processed information is provided to the user. In SchemeSathi, the DFD represents the flow of socioeconomic information from the user through data preprocessing, profile creation, scheme matching, recommendation, benefit-gap analysis, and final result generation.

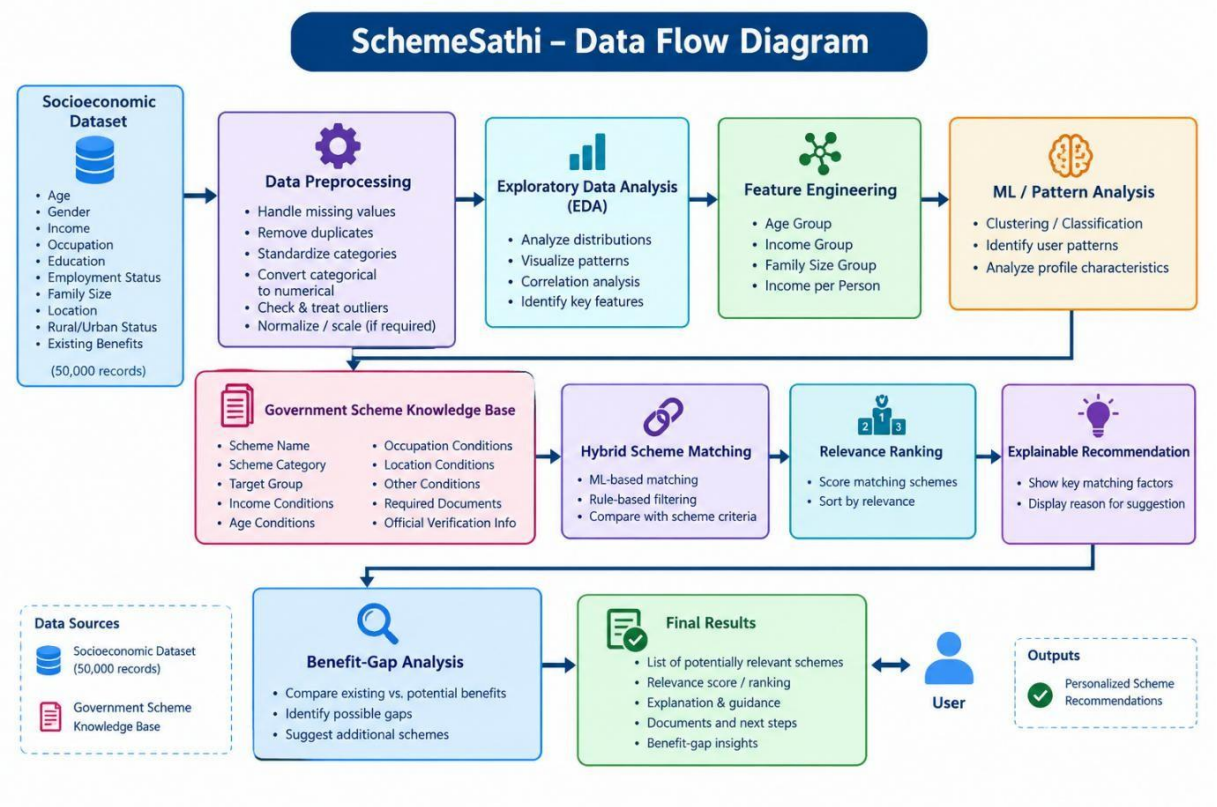


Fig 4: Data flow Diagram

3.3.1 DFD Level 0

The Level 0 DFD represents the complete SchemeSathi system as a single process. The **User/Household** provides socioeconomic information to the system. The system processes the information using the dataset, scheme knowledge base, and recommendation components. Finally, the system provides potentially relevant government schemes, explanations, possible benefit gaps, and next-step guidance to the user.

Flow:

User / Household

↓

SchemeSathi System

↓

Potentially Relevant Schemes + Explanation + Benefit Gap + Next Steps

The Level 1 DFD expands the user-data processing stage. The user provides socioeconomic information, which is passed to the data preprocessing module. The data is cleaned, transformed, and used to create a structured socioeconomic profile. The processed profile is then passed to the feature engineering and recommendation components.

Flow:

User Information
↓
Data Collection
↓
Data Preprocessing
↓
Socioeconomic Profile
↓
Feature Engineering

3.3.3 DFD Level 1 – Scheme Recommendation Module

The scheme recommendation module receives the processed socioeconomic profile and compares it with the government scheme knowledge base. Machine learning or pattern analysis and rule-based matching are performed to identify potentially relevant schemes. The matched schemes are then passed to the relevance-ranking module.

Flow:

Socioeconomic Profile
↓
ML / Pattern Analysis + Scheme Knowledge Base
↓
Rule-Based Matching
↓
Scheme Matching
↓
Relevance Ranking
↓
Potentially Relevant Schemes

3.3.4 DFD Level 1 – Explanation and Benefit Analysis Module

The explanation and benefit-analysis module receives the potentially relevant schemes and generates understandable reasons for each recommendation. If existing-benefit information is available, it is compared with the recommended schemes to identify possible benefit-access gaps. The system then generates document requirements and next-step guidance for the user.

Flow:

Potentially Relevant Schemes

↓

Why Suggested?

↓

Existing Benefits Comparison

↓

Possible Benefit Gap

↓

Documents / Next Steps

↓

Final User Results

3.4 Advantages

The proposed SchemeSathi system provides several advantages. It helps users discover government schemes based on their socioeconomic characteristics instead of requiring them to manually search through a large number of schemes. The hybrid approach combines rule-based scheme criteria with machine learning or pattern analysis to support personalized recommendations. The explainable recommendation component allows users to understand why a scheme was suggested. Benefit-gap analysis can highlight potentially relevant schemes that are not recorded among the user's existing benefits. The system can also provide information about documents and next steps, making the process easier to understand.

3.5 Requirement Specification

The requirement specification defines the hardware and software resources required to develop and operate the SchemeSathi system.

3.5.1 Hardware Requirements

The proposed system is primarily a software-based AI/ML and web application system and does not require IoT or specialized hardware components. A standard computer or laptop can be used for development and execution.

Minimum hardware requirements:

- Processor: Intel Core i3 or equivalent
- RAM: 8 GB or above
- Storage: 256 GB or above
- Display: Standard monitor
- Keyboard and mouse

- Internet connection for accessing web resources and official scheme information

3.5.2 Software Requirements

The proposed system can be developed using Python-based data science and web-development technologies.

Software requirements:

- Operating System: Windows/Linux
- Programming Language: Python
- Data Processing: Pandas, NumPy
- Machine Learning: Scikit-learn
- Data Visualization: Matplotlib / Plotly
- Backend: FastAPI or Flask
- Frontend: React / HTML / CSS
- Database: PostgreSQL or MongoDB, if required
- Development Environment: Jupyter Notebook / VS Code
- Web Browser: Google Chrome or equivalent

The proposed technology stack supports data preprocessing, machine learning, visualization, backend development, and user-interface development required for SchemeSathi.

CHAPTER 4

IMPLEMENTATION AND RESULT

4.1 Implementation

The proposed **SchemeSathi – AI-Based Government Scheme Recommendation and Eligibility Assistance System** is implemented as a Python-based AI and data analysis system. The implementation focuses on processing socioeconomic data, identifying important patterns, applying machine learning or pattern-analysis techniques, matching user profiles with government scheme criteria, ranking potentially relevant schemes, explaining the recommendations, and identifying possible benefit-access gaps. The system does not use a web application. Instead, the implementation is performed using a dataset, Python programming, data-processing libraries, machine learning techniques, and a structured government scheme knowledge base.

The overall implementation follows the methodology described in the previous chapter:

Dataset
↓
Data Preprocessing
↓
Exploratory Data Analysis
↓
Feature Engineering
↓
ML / Pattern Analysis
↓
Government Scheme Knowledge Base
↓
Hybrid Scheme Matching
↓
Relevance Ranking
↓
Explainable Recommendation
↓
Benefit-Gap Analysis
↓
Final Results

4.2 Dataset Implementation

The dataset is used as the primary input for the SchemeSathi system. It contains socioeconomic attributes that can be analyzed to understand the characteristics of individuals or households. Depending on the selected dataset, attributes may include age, gender, income, occupation, education, employment status, family size, location, rural or urban status, and existing benefits.

The dataset is loaded into the Python environment and examined to understand its structure, attributes, data types, missing values, and possible inconsistencies. The available attributes are then selected according to their relevance to socioeconomic analysis and government scheme matching.

4.2.1. Dataset Attributes and Description

S. No.	Attribute	Data Type	Description	Example
1.	Age	Numerical	Age of the individual or household membe	35
2.	Gender	Categorical	Gender of the individual	Female
3.	Income	Numerical	Income of the individual/household used for socioeconomic analysis	₹1,50,000
4.	Occupation	Categorical	Main occupation of the individual	Farmer
5.	Education	Categorical	Educational qualification	Graduate
6.	Employment Status	Categorical	Current employment condition	Employed
7.	Family Size	Numerical	Number of members in the household	5
8.	Location	Categorical	Geographic location of the individual/household	Telangana
9.	Rural/Urban Status	Categorical	Indicates whether the household is located in a rural or urban area	Rural
10.	Dataset Attributes and Description	Categorical/Text	Government benefits already received or recorded for the individual/household	Pension

Table No1:. Dataset Attributes and Description

4.3 Data Preprocessing

The collected dataset is preprocessed before applying analysis and machine learning techniques. Missing values are identified and handled using appropriate methods. Duplicate records are removed to prevent repeated observations from affecting the analysis. Inconsistent categorical values are standardized, and categorical attributes are converted into numerical representations when required.

Numerical attributes are also checked for invalid values and outliers. Where necessary, numerical features are scaled or normalized depending on the requirements of the selected machine learning algorithm. After preprocessing, the dataset is converted into a clean and consistent form for further analysis.

4.4 Exploratory Data Analysis

Exploratory Data Analysis is performed to understand the characteristics and patterns present in the socioeconomic dataset. Important attributes such as age, income, occupation, education, employment status, family size, and location are analyzed based on their availability in the selected dataset.

Statistical analysis and graphical visualization are used to study the distribution of different attributes. Bar charts, histograms, distribution plots, and correlation analysis can be used to identify relationships and patterns within the dataset.

The results obtained from EDA help in understanding the dataset and selecting useful features for the subsequent stages of the system.

4.4.1. Derived Features

S. No.	Derived Feature	Derived From	Purpose
1	**Age Group**	Age	Groups individuals according to age ranges
2	**Income Group**	Income	Converts income into meaningful income categories
3	**Family Size Group**	Family Size	Represents households as small, medium, or large
4	**Income per Person**	Income + Family Size	Represents approximate income available per household member

Table No 2:. Derived Features

4.5 Feature Engineering

Feature engineering is performed to prepare the important socioeconomic attributes for machine learning and scheme matching. Relevant attributes are selected based on their relationship with the recommendation task.

Derived features can also be created where appropriate. For example, income can be converted into meaningful income categories, and household-related attributes can be transformed into suitable representations for comparison with government scheme conditions.

The final feature set is used as an input for machine learning or pattern analysis and the scheme-matching process.

4.6 Machine Learning / Pattern Analysis

The machine learning or pattern-analysis component is implemented to analyze relationships and patterns in the socioeconomic data. The selection of the machine learning algorithm

depends on the characteristics of the final dataset and the availability of a suitable target variable.

Possible algorithms include **Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and Gradient Boosting**. The appropriate algorithm is selected after examining the dataset and determining the requirements of the prediction or analysis task.

The ML component is used to support personalized scheme analysis. It does not independently make a final decision regarding government eligibility.

4.6.1: Government Scheme Knowledge Base

S. No.	Knowledge-Base Attribute	Description
1	**Scheme Name**	Name of the government scheme
2	**Scheme Category**	Category/type of the scheme
3	**Target Group**	Intended beneficiary group
4	**Income Conditions**	Income-related conditions
5	**Age Conditions**	Age-related conditions
6	**Occupation Conditions**	Occupation-related conditions
7	**Location Conditions**	Location-related conditions
8	**Other Conditions**	Additional eligibility-related conditions
9	**Required Documents**	Documents required for verification/application
10	**Official Verification Information**	Information/source used for official verification

Table No3:.Government Scheme Knowledge Base

4.7 Government Scheme Knowledge Base

A structured knowledge base is created to store information about the government schemes considered in the project. The knowledge base contains the documented conditions and characteristics associated with each scheme.

The information may include:

- Scheme name
- Scheme category
- Target group
- Income conditions
- Age conditions

- Occupation conditions
- Location conditions
- Other relevant conditions
- Required documents
- Official verification information

The knowledge base is used by the rule-based component to compare the user's or household's socioeconomic profile with the conditions associated with different schemes.

4.8 Hybrid Scheme Matching

The hybrid scheme-matching component combines **ML/pattern analysis and rule-based reasoning**. The socioeconomic profile generated from the dataset is compared with the criteria stored in the government scheme knowledge base.

The rule-based component checks the configured conditions of each scheme, while the ML or pattern-analysis component provides additional data-driven support. The combined analysis is used to identify schemes that may be relevant to the given profile.

4.9 Relevance Ranking

When more than one scheme matches the socioeconomic profile, the relevance-ranking component is used to organize the identified schemes.

The ranking can consider factors such as the number of matching criteria, importance of the matching attributes, profile similarity, and scheme-category relevance. The resulting list allows the system to present the potentially relevant schemes in an understandable order.

The ranking represents **profile relevance**, not confirmed government eligibility.

4.10 Explainable Recommendation

The explainable recommendation component provides the reasons for identifying a scheme as potentially relevant. Instead of producing only the scheme name, the system identifies the profile attributes that contributed to the recommendation.

For example:

Scheme A – Potentially Relevant

Reasons:

- Income condition appears compatible.
- Occupation matches the configured target group.
- Age condition appears relevant.
- Location condition appears compatible.

This explanation allows the user or project evaluator to understand how the recommendation was generated. Feature-importance methods can also be applied to the ML component when appropriate.

4.11 Benefit-Gap Analysis

The benefit-gap analysis component compares the potentially relevant schemes with the benefits already recorded for an individual or household, when such information is available in the dataset.

For example:

4.11.1. Benefit-Gap Analysis

Scheme	Profile Match	Existing Benefit
Scheme A	Potentially Relevant	Recorded
Scheme B	Potentially Relevant	Not Recorded
Scheme C	Not Matched	—

Table No4:. Benefit-Gap Analysis

If a potentially relevant scheme is not recorded among the existing benefits, it is identified as a **possible benefit-access gap**.

The system does not interpret this as proof that the person is eligible or has been denied the benefit. It only indicates that the scheme may need to be checked.

4.12 Document and Next-Step Guidance

The system provides additional information associated with the identified schemes. This may include eligibility conditions to verify, required documents, official sources, and suggested next steps.

This component helps users understand what information needs to be verified before proceeding with a government scheme application.

4.13 Implementation Environment

The SchemeSathi system is implemented using Python and commonly used data science and machine learning libraries.

The implementation environment includes:

- **Programming Language:** Python
- **Data Processing:** Pandas, NumPy
- **Machine Learning:** Scikit-learn
- **Data Visualization:** Matplotlib / Plotly
- **Development Environment:** Jupyter Notebook / VS Code
- **Dataset:** Socioeconomic dataset

- **Knowledge Base:** Structured government scheme criteria

No hardware components or web application are required for the implementation of the proposed system.

4.14 System Output

The final output of SchemeSathi consists of potentially relevant government schemes identified from the socioeconomic information.

The output includes:

1. Socioeconomic profile summary
2. Potentially relevant schemes
3. Relevance information
4. Reasons for recommendation
5. Existing benefit information, where available
6. Possible benefit-access gaps
7. Required documents
8. Conditions to verify
9. Official source information
10. Suggested next steps

The output allows the user to understand which schemes may be relevant and why they were identified.

4.15 Results

The implemented system demonstrates the use of socioeconomic data for personalized government scheme discovery. After preprocessing and analyzing the dataset, relevant socioeconomic characteristics are identified and used in the scheme-matching process. The hybrid approach combines data-driven analysis with configured scheme criteria to identify potentially relevant government schemes.

The explainable recommendation component provides the factors responsible for each recommendation, while the benefit-gap component can identify potentially relevant schemes that are not recorded among the existing benefits when such information is available.

The final numerical results depend on the actual dataset and the machine learning model selected during implementation. Therefore, model performance values such as accuracy, precision, recall, and F1-score should be added after the final model is trained and evaluated.

4.16 Model Evaluation

The performance of the machine learning component is evaluated according to the type of problem supported by the selected dataset.

For a supervised classification problem, the following evaluation measures can be used:

- **Accuracy**
- **Precision**
- **Recall**
- **F1-Score**
- **Confusion Matrix**

For recommendation or ranking-based analysis, suitable recommendation metrics can be used. The rule-based scheme-matching component can also be tested using manually verified sample cases to check whether the configured criteria produce the expected matching results.

The final evaluation values should be obtained from the actual implementation rather than using assumed or sample values.

CHAPTER 5

DISCUSSION AND CONCLUSION

5.1 Key Findings

The project successfully developed a structured approach for **Personalized Government Scheme Discovery and Eligibility Assistance Using Socioeconomic Data**. The key finding is that socioeconomic information can be processed and analyzed systematically to identify government schemes that may be relevant to an individual or household.

The dataset was first examined to understand its columns and the information available for analysis. The data was then cleaned by handling missing values, duplicate records, inconsistent values, and categorical attributes where required. Exploratory Data Analysis (EDA) helped identify important patterns and relationships among socioeconomic factors such as **age, income, occupation, education, employment status, family size, and location**.

The prepared features were then used for **Machine Learning (ML)** to identify useful socioeconomic patterns. The ML stage provides analytical support for the recommendation process rather than making a final legal eligibility decision.

The **scheme-matching module** combines the user's socioeconomic profile with predefined government-scheme criteria. This allows the system to identify **potentially relevant schemes** based on the available profile information. The approach provides more personalized results than simply presenting a general list of government schemes.

The project also provides an explainable recommendation process in which users can understand **why a particular scheme was suggested**, based on matching factors such as income, age, occupation, education, or location. Finally, the developed application presents the recommended schemes and relevant eligibility-assistance information through a user-friendly interface.

Overall, the project demonstrates the feasibility of using **socioeconomic data, EDA, Machine Learning, and scheme-based matching** to build a personalized government scheme discovery system. The system assists users in identifying programmes that may be relevant to them while emphasizing that final eligibility must be verified according to the latest official government requirements.

5.2 Git Hub Link of the Project:

5.3 Video Recording of Project:

https://drive.google.com/file/d/1bmIxbsxgwkSzfimx19Zyj6h20M6cDpA1/view?usp=sharing

5.4 Limitations

The current SchemeSathi system has some limitations. The accuracy and usefulness of the recommendations depend on the **quality, completeness, and relevance of the socioeconomic dataset and government scheme information** used by the system. Missing, outdated, or inconsistent user information may affect the recommendations.

The system provides **potentially relevant schemes rather than final eligibility decisions**, because actual eligibility depends on the latest rules, conditions, documents, and verification procedures defined by the respective government authorities. Changes in scheme guidelines may therefore require the scheme knowledge base to be updated.

Another limitation is that the performance of the machine learning component depends on the availability of suitable training data and meaningful target labels. If the dataset does not contain sufficient scheme-related labels or representative socioeconomic information, the ML model may have limited predictive capability.

The current system also focuses mainly on the socioeconomic and demographic information available in the selected dataset. Additional factors that may influence scheme eligibility may not be represented. Therefore, the recommendations should be treated as an assistance mechanism rather than a replacement for official eligibility verification.

5.5 Future Work

The system can be improved in several ways in future work. A larger and more diverse socioeconomic dataset can be used to improve the reliability of pattern analysis and machine learning models. More comprehensive and regularly updated government scheme information can also be incorporated into the scheme knowledge base.

Future versions can include **automatic updating of scheme criteria** from verified official government sources so that changes in eligibility conditions, benefits, and required documents can be reflected more quickly.

The machine learning component can also be improved by testing and comparing different algorithms and selecting the most suitable model based on appropriate evaluation metrics. Advanced explainable AI techniques can be incorporated to provide more detailed explanations of recommendation results.

Future work can also extend the benefit-gap analysis to consider additional household characteristics and previously received benefits. A larger set of schemes and more detailed document and application guidance can further improve the usefulness of the system.

Finally, the system can be evaluated using a larger, manually verified test dataset and real-world user scenarios to measure the quality of scheme recommendations and identify areas for further improvement.

5.6 CONCLUSION

The **AI-Based Government Scheme Recommendation and Eligibility Assistance System (SchemeSathi)** provides a structured approach for identifying government schemes that may be relevant to users based on their socioeconomic information. The system combines **data preprocessing, exploratory data analysis, feature engineering, machine learning/pattern analysis, and rule-based scheme matching** to analyze user profiles and recommend potentially relevant schemes.

The hybrid approach helps connect user-specific characteristics with predefined government scheme criteria. The system also provides **explainable recommendations** by showing the factors that contributed to a scheme being suggested. In addition, the **benefit-gap analysis** compares potentially relevant schemes with the benefits already recorded for the user, helping identify possible areas where additional benefits may need to be explored.

Overall, SchemeSathi demonstrates how **AI and data analytics can support personalized government scheme discovery** in a structured and understandable manner. The system is intended to assist users in finding relevant schemes and understanding the information and documents required for further verification. Since government eligibility rules may change and final eligibility is determined by the respective authorities, the system provides **potential recommendations rather than guaranteed eligibility decisions**.

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